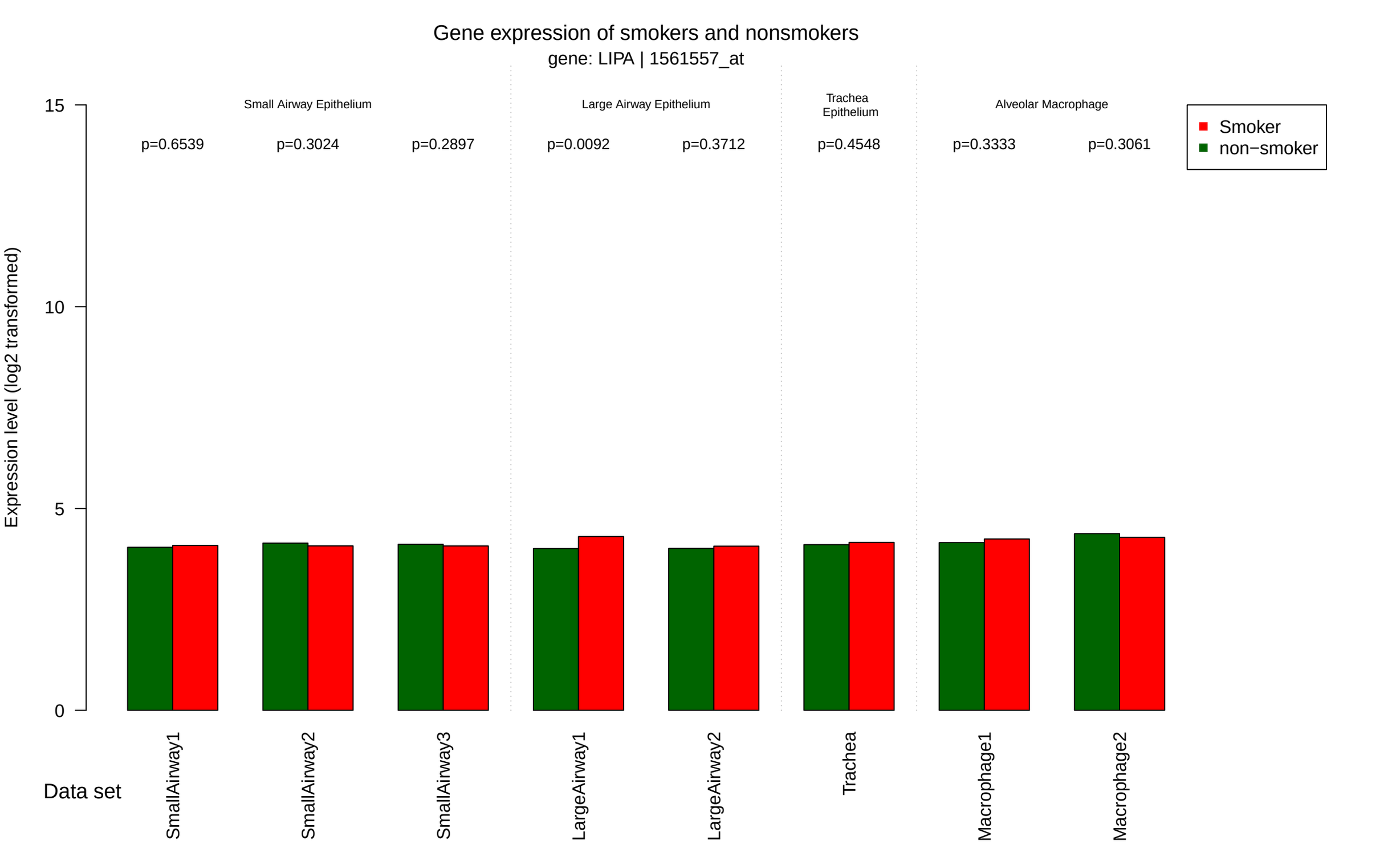
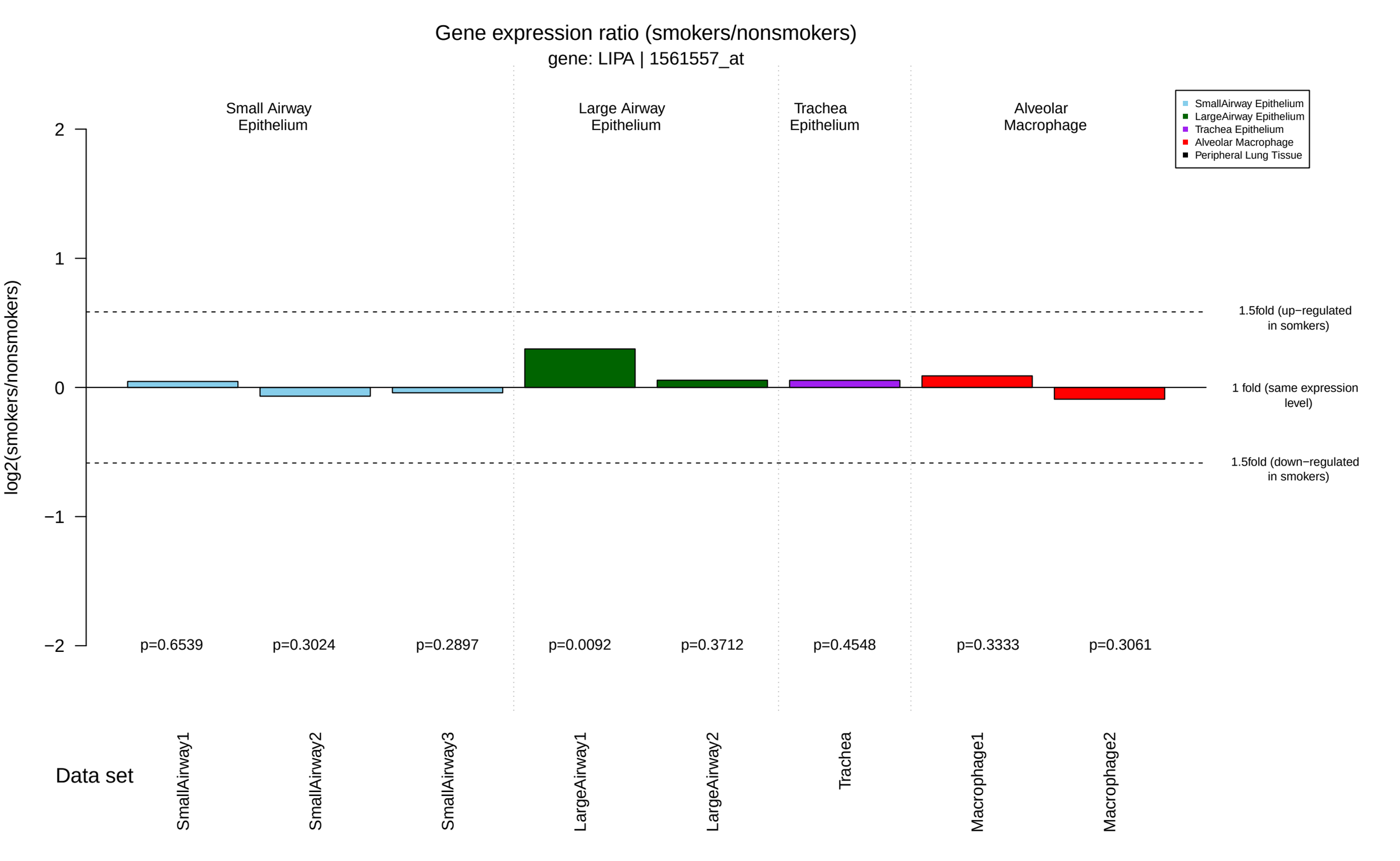
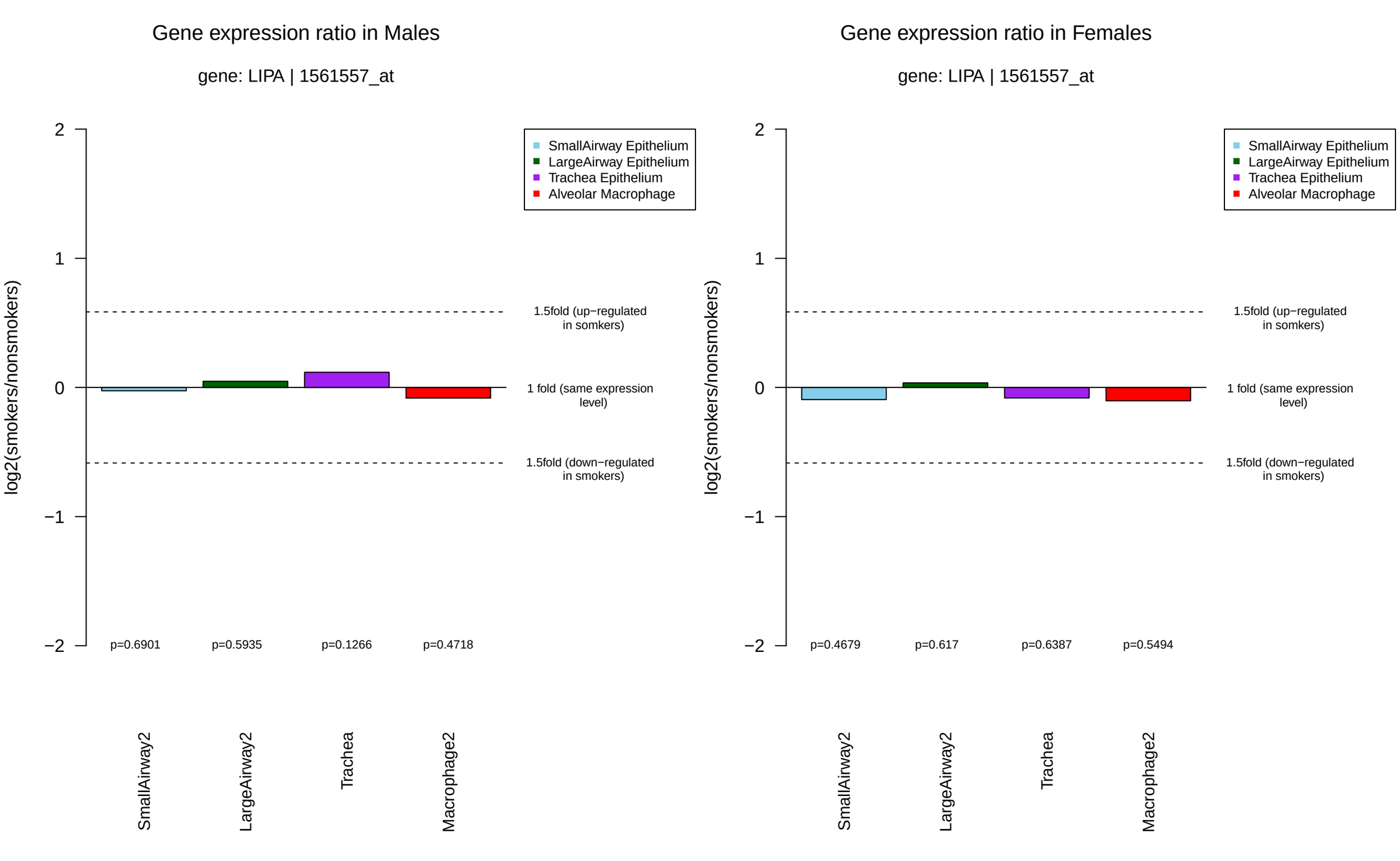
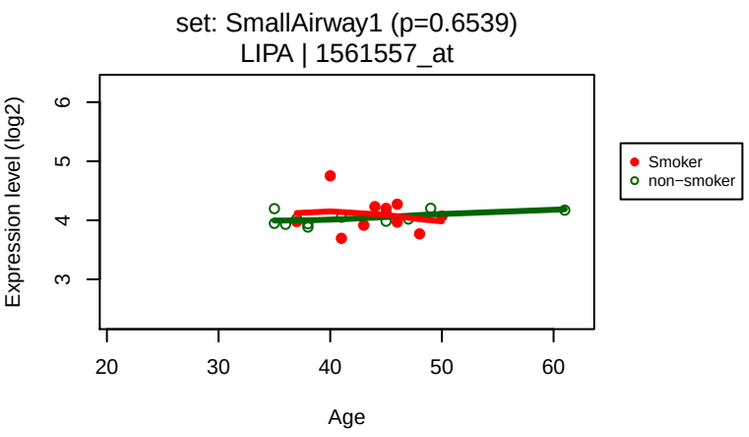
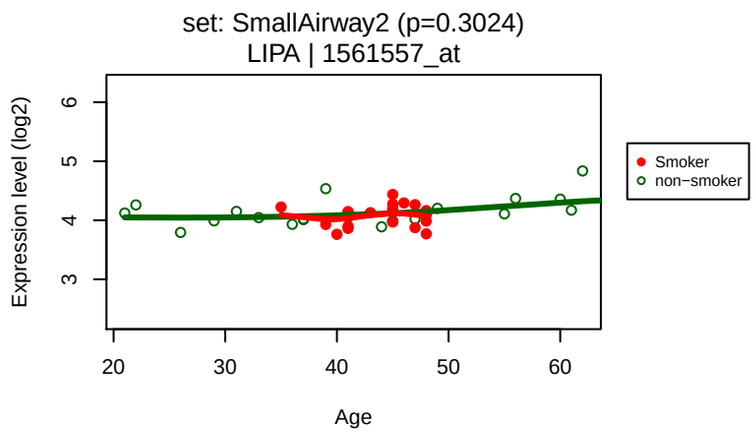
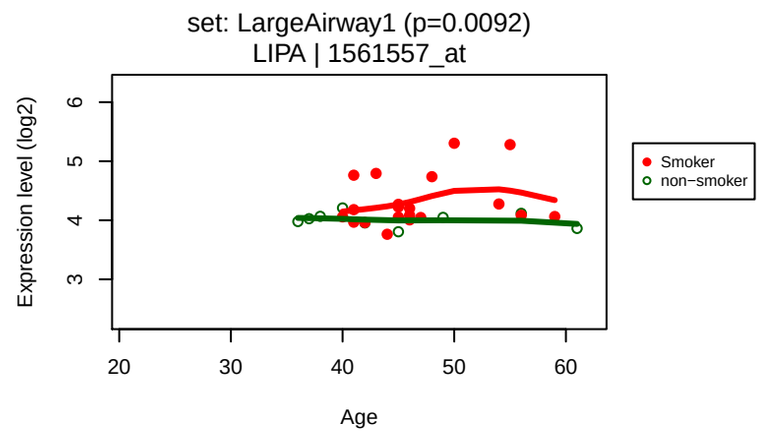

Gene expression of smokers and nonsmokers
gene: LIPA | 1561557_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6539
p=0.3024
p=0.2897
p=0.0092
p=0.3712
p=0.4548
p=0.3333
p=0.3061
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

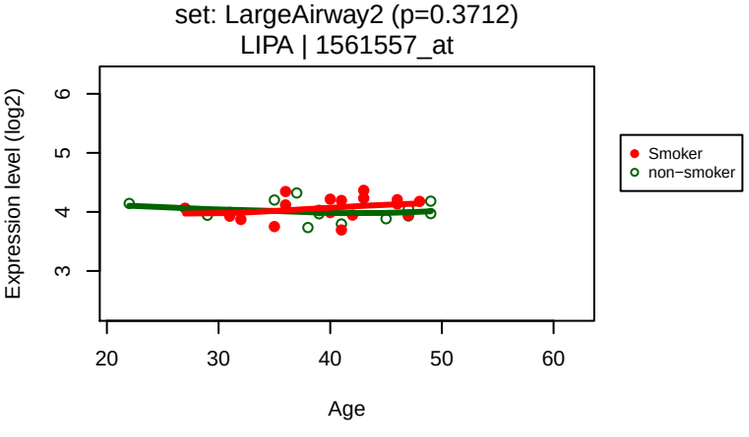

Gene expression ratio (smokers/nonsmokers)
gene: LIPA | 1561557_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6539
p=0.3024
p=0.2897
p=0.0092
p=0.3712
p=0.4548
p=0.3333
p=0.3061
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

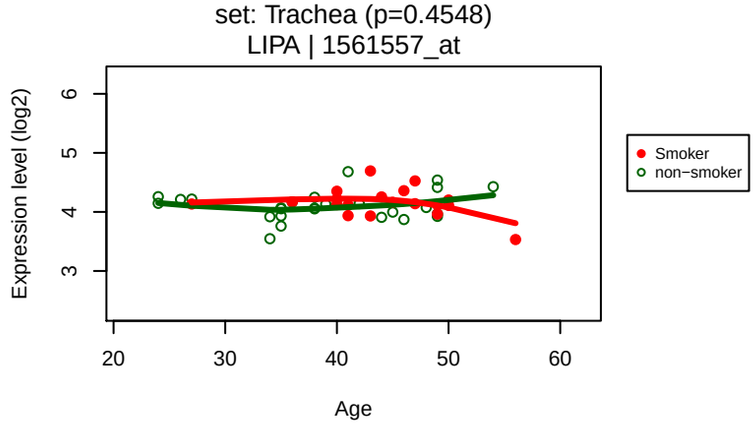

Gene expression ratio in Males
gene: LIPA | 1561557_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6901
p=0.5935
p=0.1266
p=0.4718
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LIPA | 1561557_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4679
p=0.617
p=0.6387
p=0.5494
SmallAirway2
LargeAirway2
Trachea
Macrophage2

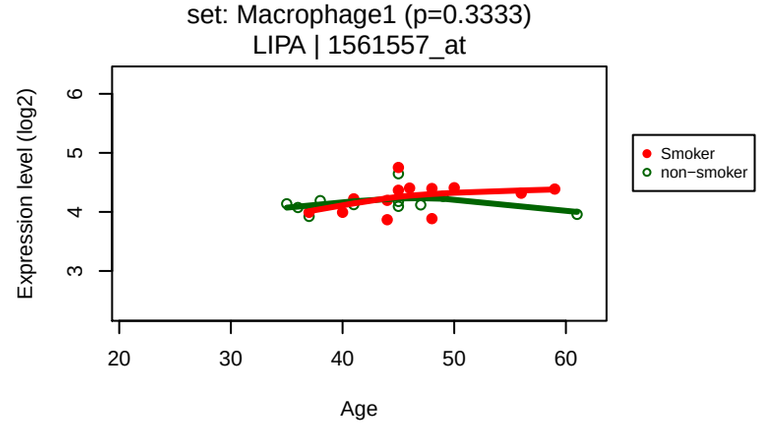

set: SmallAirway1 (p=0.6539)
LIPA | 1561557_at
Expression level (log2)
Age
Smoker
non−smoker

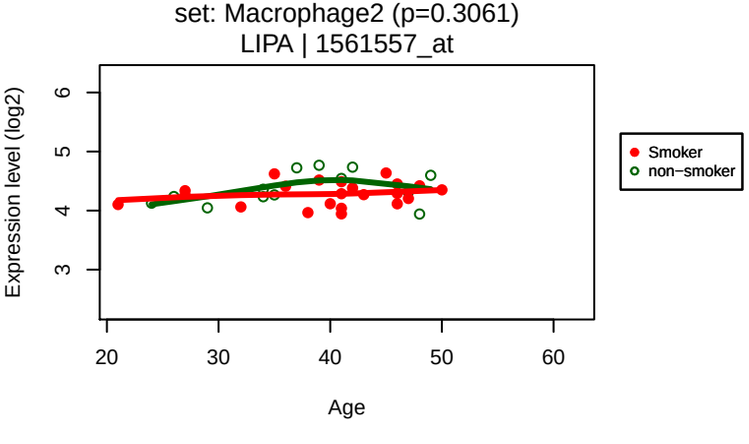

set: SmallAirway2 (p=0.3024)
LIPA | 1561557_at
Expression level (log2)
Age
Smoker
non−smoker

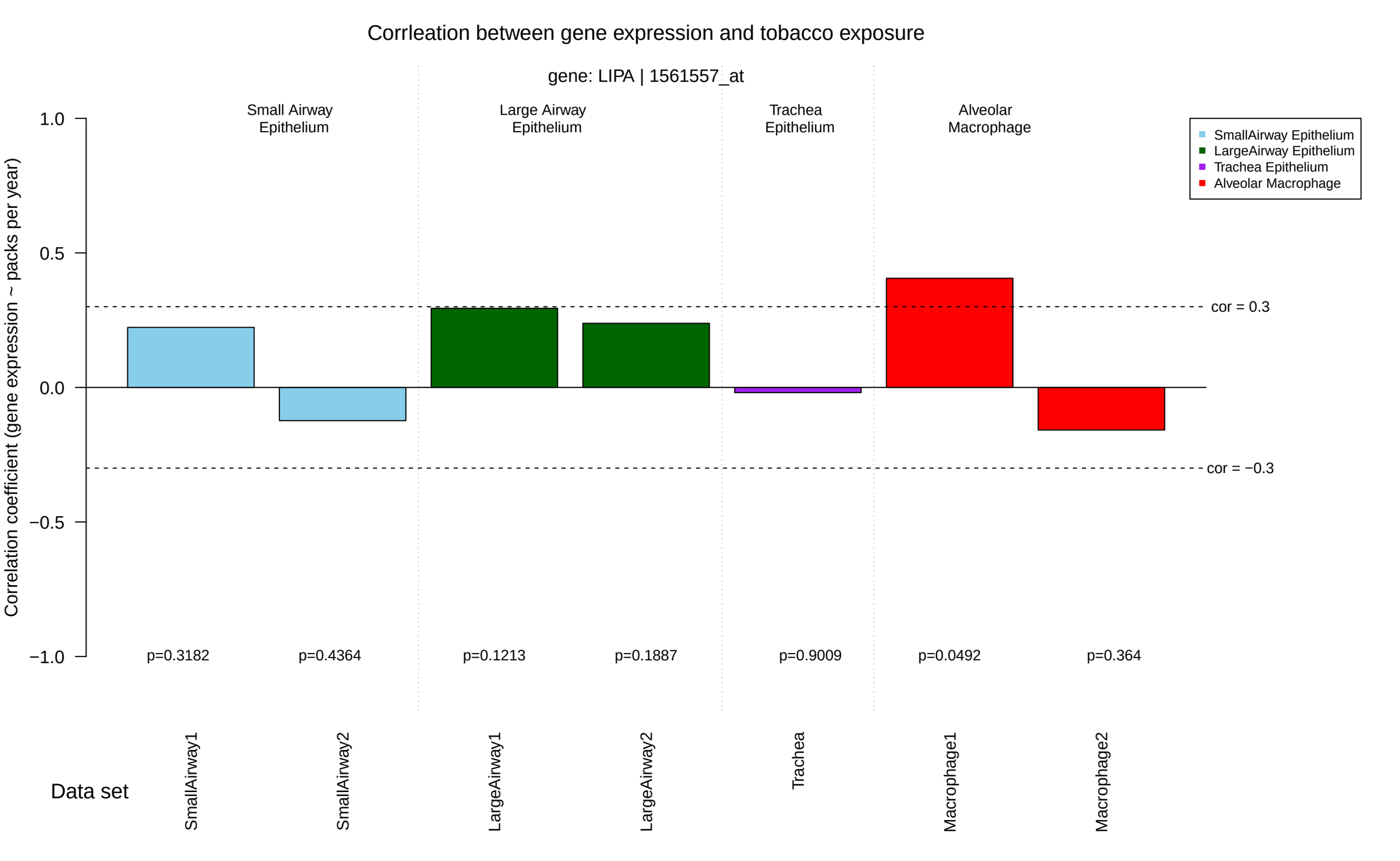

set: LargeAirway1 (p=0.0092)
LIPA | 1561557_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3712)
LIPA | 1561557_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4548)
LIPA | 1561557_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3333)
LIPA | 1561557_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3061)
LIPA | 1561557_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LIPA | 1561557_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3182
p=0.4364
p=0.1213
p=0.1887
p=0.9009
p=0.0492
p=0.364
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2